

Aggressive trait selection influences life-history trade-offs, favoring survival over reproductive success in *Drosophila melanogaster* males

Reviewed Preprint

v1 • October 21, 2025

Not revised

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eLife Assessment

This **valuable** study addresses the effects of selection on aggression on fitness and life-history trade-offs in *Drosophila melanogaster*. However, the evidence presented is **incomplete** and does not support the claims proposed in the study of increased survival of highly aggressive males at the expense of reproductive success and shorter mating duration. The main limitation of the study is the choice to use males from only one aggressive *Drosophila* line in combination with CantonS females, that do not allow disambiguation between nonaggression-related factors, such as hybrid vigor and aggression-related factors influencing mating and lifespan.

<https://doi.org/10.7554/eLife.107559.1.sa3>

Abstract

Life history traits are all essential traits linked to development, survival and reproduction, directly influencing individual fitness. Limited environmental or physiological resources force organisms to balance competing demands, leading to fundamental trade-offs, particularly between survival and reproduction. Male aggression often enhances reproductive success by improving resource acquisition and mate competition. However, the broader impact of high levels of aggression on individual overall fitness and life-history trade-offs remains poorly understood. To address this, we used the *Drosophila melanogaster* *Bully* line, derived from wild-type flies (*Canton-S* line) through artificial selection for high levels of male aggression, and assess whether increased aggressiveness influences male life-history traits. Our results reveal that increased male aggressiveness shifts the balance of a major life-history trade-off, favoring survival over reproductive success. *Bully* males exhibited lower mating success, shorter mating duration and less effective chemical mate-guarding. However, this was traded-off with increased lifespan, enabling more reproductive opportunities. We also uncover potential mechanistic underpinnings of the reproductive deficits. *Bully* males showed differences in their cuticular hydrocarbon (CHCs) profiles and

transferred lower levels of cVA to females, a mate-guarding pheromone, thereby weakening their post-mating strategy. Overall, our findings demonstrate that selection for male aggression profoundly influence a key life-history trade-off highlighting aggression as an evolutionary force shaping adaptation of life-history traits and providing a foundation for future genetic and mechanistic studies.

Introduction

Life history traits, such as development time, fertility, survival, and reproductive success, are key phenotypic components that directly influence individual fitness (Braendle and Paaby, 2024 [↗](#); Roff, 1992 [↗](#); Stearns, 1992 [↗](#)). These traits often face trade-offs: investing in one can compromise another (Fisher, 1930 [↗](#); Stearns, 1989 [↗](#); Williams, 1966 [↗](#)). For instance, high reproductive effort may reduce survival due to the energetic costs of gamete production or increased mortality associated with mating behaviors or competition (Flatt, 2011 [↗](#); 2020 [↗](#)). Trade-offs also emerge between pre- and post-copulatory strategies (Simmons, 2001 [↗](#); Warner et al., 1995 [↗](#)). Males prioritizing pre-copulatory investment, such as aggressive competition for mates, may have limited resources for post-copulatory mechanisms, including sperm competition or mate guarding (Andersson, 1994 [↗](#); Brennan and Orbach, 2021 [↗](#)). Understanding how such trade-offs are shaped by natural selection offers insight into how organisms optimize fitness in competitive social environments (Roff, 1992 [↗](#); Stearns, 1976 [↗](#)).

The fruit fly *Drosophila melanogaster* (*D. mel.*), provides a powerful model for studying life-history trade-offs, with well-characterized developmental, behavioral, and reproductive traits (Flatt, 2020 [↗](#)). These traits are shaped by both genetic and environmental factors and can evolve under ecological pressures. Faster development, earlier reproduction and high mating rate, for instance, are advantageous in changing environments but come at the cost of reduced longevity and fecundity (Partridge and Farquhar, 1981 [↗](#); Stearns, 2000 [↗](#)). Experimental manipulations of environmental conditions such as temperature and food availability reveal how external factors shape trade-offs between survival and reproduction (Chippindale et al., 1996 [↗](#); Marshall and Sinclair, 2010 [↗](#)).

Beyond development and fecundity, *D. mel.* serves as a model to study a wide range of social behaviors, including aggregation, aggression, and courtship, that are largely mediated by chemical communication (Kim et al., 2017 [↗](#); Kohl et al., 2015 [↗](#)). Cuticular hydrocarbons (CHCs) and pheromones play diverse roles in social interaction: 7-Tricosene (7-T), a male-specific compound, increases female receptivity and modulates aggression (Grillet et al., 2006 [↗](#); Wang et al., 2011 [↗](#)); cis-vaccenyl acetate (cVA), a male-transferred pheromone, attracts females, repels other males, and promotes aggregation (Kurtovic et al., 2007 [↗](#); Laturney and Billeter, 2016 [↗](#)); and the female-specific hydrocarbon 7,11-heptacosadiene (7,11-HD) strongly elicits male courtship and supports species-specific mate recognition (Toda et al., 2012 [↗](#)).

Among these interactions, male-male competition plays a central role in shaping reproductive strategies. Pre-copulatory competition involves resource defense, courtship displays, and aggressive interactions to secure mating opportunities. Courtship in *Drosophila* includes stereotyped behaviors such as orientation, wing extensions (producing courtship song), tapping, licking, and attempted copulation (Yamamoto and Koganezawa, 2013 [↗](#)). Successful courtship displays lead to female acceptance, outcompeting other males that fail to meet the female's preferences (Baxter et al., 2018 [↗](#); Hindmarsh Sten et al., 2025 [↗](#)). Aggression intensifies this competition: more aggressive or dominant males often monopolize access to females (Dow and von Schilcher, 1975 [↗](#); Filice and Dukas, 2019 [↗](#); Gao et al., 2024 [↗](#); Nandy et al., 2016 [↗](#); Prunier and Trannoy, 2024 [↗](#)). Aggressive behaviors range from wing threats and lunging to high-intensity physical confrontations like tussling and boxing (Chen et al., 2002 [↗](#)). These behaviors promote

territorial dominance and enhance mating success (Gao et al., 2024 [DOI](#)), although subordinate males may also achieve reproductive success through first access to females and prolonged copulation (Filice and Dukas, 2019 [DOI](#)).

Post-copulatory competition further shapes male reproductive success. After mating, males use multiple strategies to ensure paternity, including the transfer of seminal fluid proteins (SFPs) including Sex Peptide (SP) and Accessory gland proteins (Acps), which promote sperm storage, stimulate oviposition, and reduce female receptivity to further mating (Chapman et al., 2003 [DOI](#); Fricke et al., 2009 [DOI](#)). These factors also contribute to the formation of a mating plug that retains sperm, acting as a passive mate-guarding strategy (Avila et al., 2011 [DOI](#); Dunham and Rudolf, 2009 [DOI](#)). In parallel, chemical mate guarding via pheromone transfer (e.g. cVA and 7-T) reduce female attractiveness and their likelihood of re-mating (Billeter et al., 2009 [DOI](#); Gillott, 2003 [DOI](#); Laturney and Billeter, 2016 [DOI](#); Miyamoto and Amrein, 2008 [DOI](#); Scott, 1986 [DOI](#); Zawistowski and Richmond, 1986 [DOI](#)). Male reproductive behavior is plastic: in the presence of rivals, males extend their mating duration (Bretman et al., 2009 [DOI](#); Bretman et al., 2013 [DOI](#); Kim et al., 2013 [DOI](#)), increase the number of sperm in their ejaculate (Garbaczewska et al., 2013 [DOI](#)), adapt seminal fluids content in their ejaculate (Hopkins et al., 2019 [DOI](#)), and present higher levels of aggression toward opponents to guard their mates (Baxter et al., 2015 [DOI](#); Bretman et al., 2011 [DOI](#); Dore et al., 2021 [DOI](#); Mazzi et al., 2009 [DOI](#)). Together, this demonstrate that male-male aggression influences access to mates and reproductive outcomes through both pre- and post-copulatory competition.

While aggression can enhance male mating success in natural settings, the relationship between aggression and reproductive fitness is not always straightforward. In a previous study, Penn et al. (Penn et al., 2010 [DOI](#)) developed “*Bully*” males through 37 generations of selection for aggression originating from a *Canton-S* (Cs) background. These hyper-aggressive males frequently escalated to boxing, the most intense aggression phenotype (Chowdhury et al., 2017 [DOI](#)), but showed reduced reproductive success with virgin females compared to unselected control males (Penn et al., 2010 [DOI](#)). This suggests that although aggression may enhance competitive ability, it can also incur fitness costs. This raises important questions about the role of aggression in shaping life-history trade-offs, such as: to what extent does aggression influence life history traits related to reproduction, and does it affect the balance reproduction vs survival?

To address these questions, we investigated whether *D. mel.* males from the *Bully* line exhibit changes in life-history trade-offs compared to the unselected starting wild-type flies (Cs lines). We found that hyper-aggressive *Bully* males show reduced reproductive success and shorter mating duration across their lifespan, indicating lower pre-copulatory success. They also display less effective mate-guarding behavior, indicating a weaker post-copulatory strategy. However, this poor pre- and post-copulatory success was traded-off with an increased survival, which is accompanied by late-life reproductive success. We then provided a mechanistic basis for the reduced copulatory success of *Bully* males. They showed distinct CHCs profiles compared to Cs males, which may contribute to their lower mating success. Moreover, they transferred lower levels of cVA, a mate-guarding pheromone, to females, potentially decreasing efficiency of their post-mating strategy. Our findings underscore the complex interplay between aggression, reproductive behavior, survival, and chemical signaling, providing insights into the evolutionary dynamics of life-history trade-offs in competitive social contexts.

Results

Highly aggressive males exhibit reduced mating success and have shorter mating durations

To test the hypothesis that selection for male aggressive traits may confer mating advantages, we performed courtship competition assays in which *Cs* and *Bully* males competed for mating with virgin *Cs* females. No significant mating advantage was observed for *Bully* males (**Figure 1A** [↗](#)). We then assessed the reproductive capacity of *Cs* and *Bully* males in no-choice assays by pairing each male with a virgin *Cs* female for 30 minutes. Mating success was significantly lower in *Bully* males compared to what was observed with *Cs* males (**Figure 1B** [↗](#)), despite no differences in the percentage of Unilateral Wing Extension (UWE), courtship latency or mating latency, when they succeed mating (**Figure 1C-E** [↗](#)), suggesting intact courtship and mating motivation. These results were replicated using an independently selected *Bully* line (referred to as *Bully B* (Penn et al., 2010 [↗](#))), which also showed reduced mating success and increased aggression compared to *Cs* (**Figure 1-figure supplement 1A, C-E** [↗](#)). This indicates that two independent selections for male aggressive traits had simultaneously reduced males mating success.

Differences between *Cs* and *Bully* extended to mating duration (MD) where *Bully* males mated 25% shorter than *Cs* males (**Figure 1F** [↗](#)), a difference also observed in *Bully B* males (**Figure 1-figure supplement 1B** [↗](#)). This MD reduction persisted when using decapitated females (**Figure 1G** [↗](#)), ruling out female-driven termination and implicating male-intrinsic factors.

As males raised with other males have lengthened MD (Bretman et al., 2013 [↗](#); Kim et al., 2013 [↗](#)), we compared singly and group-housed males for their MD. We found that both *Cs* and *Bully* group-housed males exhibited significantly extended mating durations compared to single-housed males (**Figure 1H** [↗](#)). However, the MD in *Bully* males remained shorter compared to *Cs* males in group-housed condition (**Figure 1H** [↗](#)). This indicates that *Cs* and *Bully* males respond equally to early social experience and hence do not display gene-by-environment interactions.

Together, these findings show that selection for male aggression reduces male mating success and mating duration, key traits of pre-copulatory reproductive performance. This supports our main hypothesis that selection for aggressive traits influences either or both pre- and post-copulatory strategies.

Hyper-aggressive males display reduced mate-guarding efficiency, without compromising female fertility

Given that *Bully* males have shorter MD and reduced mating success, we hypothesized they would compensate this poor pre-copulatory success with more efficient post-mating strategies. After mating, males typically enter a refractory period during which they are no longer aroused and do not court the females they have just mated with (Lynn et al., 2024 [↗](#)). We first tested the hypothesis that *Bully* males would have a shorter refractory period than *Cs* males, allowing them to potentially mate with more females. To investigate this, we paired *Cs* and *Bully* males with virgin *Cs* females, allowed them to mate, and measured the percentage of UWE directed by these males at the female they just finished mating with over a 5-minutes post-mating period. While *Cs* males performed minimal UWE towards just-mated females (**Figure 2A** [↗](#)), *Bully* males displayed a significantly higher percentage of UWE in this context compared to *Cs* males (**Figure 2A** [↗](#)), suggesting that *Bully* males have a shorter refractory period than *Cs* males to court mated females. This effect persisted when males were swapped after mating, with *Bully* males showing increased UWE towards *Cs*-mated females (**Figure 2B-B** [↗](#)). This indicates that *Bully* males present a lower refractory period to court any mated females, allowing them to potentially have higher remating rate compared to *Cs* males. To test this, we conducted behavioral assays in which a single *Cs* or *Bully* male was consecutively paired with 6 novel virgin females over a 3h period and scored the number of mating. Contrary to our hypothesis, *Bully* males did not show higher remating rates, despite consistently shorter MD (**Figures 2C-D** [↗](#)). This suggests that, despite shorter refractory

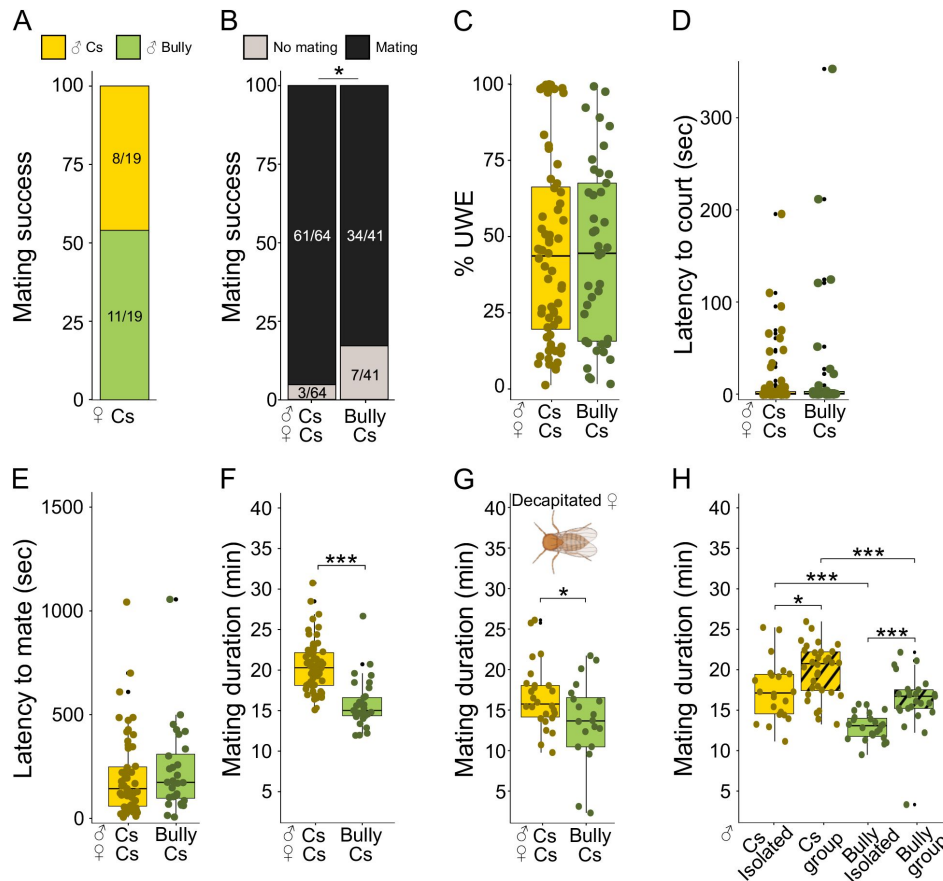


Figure 1.

Bully males exhibit reduced mating success and allocate less time to mating.

(A) In competitive courtship assays, the mating success of *Cs* and *Bully* males were scored when paired time with a single virgin *Cs* female. (B) In non-competitive courtship assays, the mating success of *Cs* and *Bully* males paired with a virgin *Cs* female were analyzed. (C) Proportion of time spent by *Cs* and *Bully* males doing Unilateral Wing Extension (UWE) toward a *Cs* virgin female in non-competitive courtship assays. (D-E) Latency to court (time difference between first interaction and first UWE) and to mate (time difference between first interaction and start of the mating) of *Cs* and *Bully* males paired with a virgin *Cs* female. (F) Mating duration of *Cs* and *Bully* males paired with a virgin *Cs* female or (G) a virgin *Cs* decapitated female. (H) Mating duration of *Cs* and *Bully* males that were previously raised in group of 10 males or in social isolation. For all graphs, stars indicate significant differences (* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$) and statistical details are given in **Figure 1-table supplement 2**.

period and reduced MD, *Bully* males did not achieve a higher remating success rate. As mating accompanies ejaculate transfer and incurs a reduction in sperm and seminal fluid reserve in *D. mel.* males (Hihara, 1981 [↗](#); Sirot et al., 2009 [↗](#)), we hypothesized that *Bully* males might retain more sperm, potentially enabling them to fertilize more females sequentially within a short period. We thus analyzed female fertility after each consecutive mating and found no significant difference between *Bully* and *Cs* males (**Figure 2E** [↗](#)). Females that mated with *Bully* males remained as fertile as those mated with *Cs* males (**Figure 2E** [↗](#)). Together, our results indicate that while *Bully* males present shorter refractory period after mating, but their remating rate and ability to fertilize females remain comparable to those of *Cs* males.

Due to a shorter MD in *Bully* males, we reasoned that a lower quantity of CHCs, acting as mate-guarding pheromones, may be transferred to the females' body, making *Cs* females that mated with *Bully* males more attractive to other males compared to those mated with *Cs* males. To assess female attractiveness independently of any confounding effects of female receptivity, we performed preference assays for decapitated females, who lack receptivity. *Cs* females were first mated with either *Cs* or *Bully* males, then decapitated and simultaneously presented to naïve males in a binary choice assay (**Figure 2F** [↗](#)). Both *Cs* and *Bully* males preferentially courted decapitated females that had previously mated with *Bully* males (**Figure 2F** [↗](#), green side), showing that females mated with *Bully* males remain more attractive than those mated with *Cs* males. In addition to mate guarding pheromones, males transfer seminal fluids that affect long term female receptivity, which may differ between *Bully* and *Cs* males. We therefore conducted mating assays in which virgin *Cs* females were initially mated with either *Cs* or *Bully* males. Several days later, these mated females were paired with naïve virgin *Cs* males, and the proportion of females that remated was analyzed (**Figure 2G** [↗](#)). Females previously mated with *Cs* males, remated significantly less than females that had mated with *Bully* males (**Figure 2G** [↗](#), green), further supporting weaker mate-guarding by *Bully* males.

We conclude that *Bully* males fail to offset their poor pre-copulatory success with more efficient post-mating strategies. Their mates present a higher remating rate, likely due to reduced transfer of mate-guarding pheromones and seminal fluid. Thus, selection for aggressive traits compromises both pre- and post-copulatory male performance.

Hyper-aggressive males present differences in their CHCs profiles and transfer lower quantities of mate-guarding pheromones to females

To identify mechanisms underlying reduced mating success (**Figure 1** [↗](#)) and mate-guarding (**Figure 2F-G** [↗](#)) in *Bully* males, we reasoned that these could be attributable to differences in males and females CHCs profiles. To test this, we quantified the CHCs profiles of naïve *Cs* and *Bully* males. PCA analysis revealed distinct CHCs profiles between naïve *Cs* and *Bully* males (**Figure 3A** [↗](#)). Comparing individual compounds of naïve *Bully* males relatively to *Cs* ones, we found that *Bully* males exhibited significantly higher levels of alkanes, monoenes, and methyl-alkanes (**Figure 3B** [↗](#), **Table 1** [↗](#)). These differences may contribute to modulating female attractiveness toward males.

Our results suggest that the higher remating rate in females mated with *Bully* males (**Figure 2G** [↗](#)) may be due to increased attractiveness compared to females mated with *Cs* males (**Figure 2F** [↗](#)), a process influenced by pheromones transferred from males to females such as the blend of cVA/7-T (Laturney and Billeter, 2016 [↗](#)). To test this hypothesis, we extracted CHCs from the cuticles of *Cs* virgin and *Cs* females mated to either a *Cs* or *Bully* males and compared their CHCs profiles. As a first step, PCA analysis showed that *Cs* virgin females exhibited distinct CHCs profiles from those of both *Cs* and *Bully*-mated females (**Figure 3C** [↗](#)). The first (PC1) and the second (PC2) dimensions explain 80.7% and 11.1% of the variance in CHCs profiles, respectively. High $\cos^2$ values indicate a strong correlation between CHCs compounds and the principal components, suggesting that the

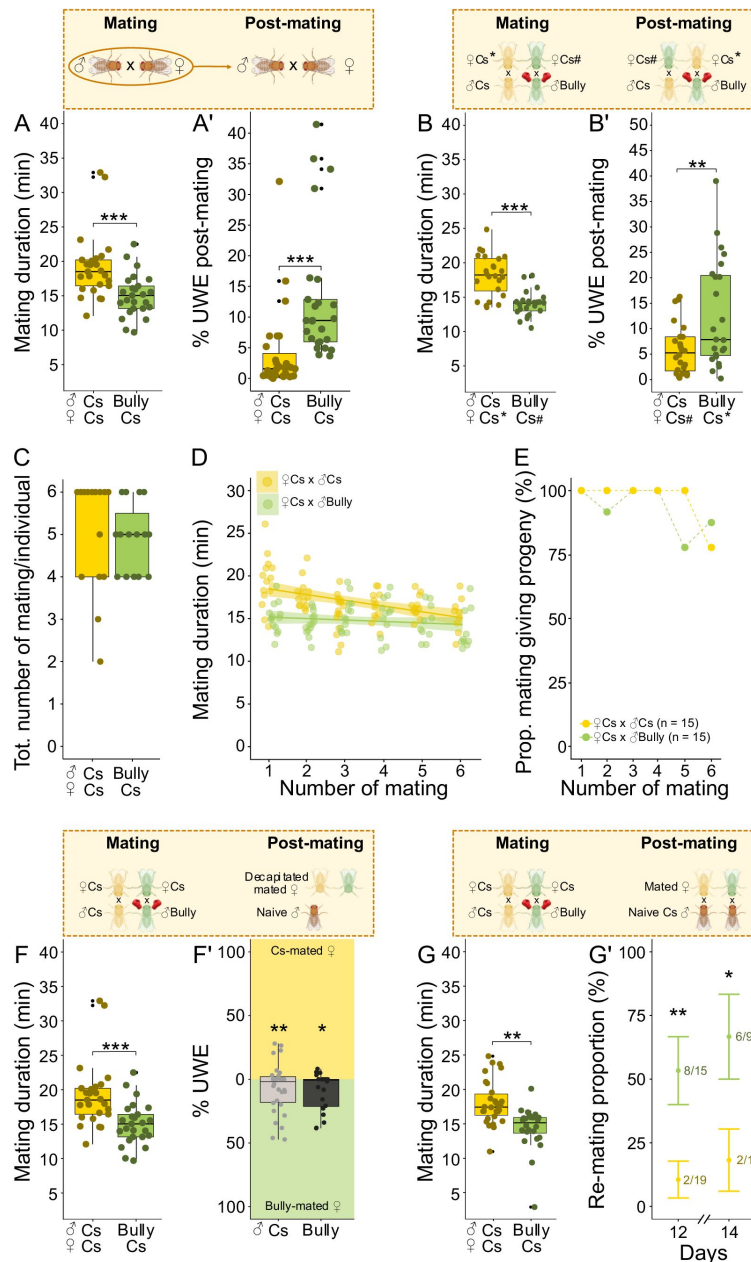


Figure 2.

Hyper-aggressive males exhibit lower mate-guarding efficiency, yet this does not affect female fertility.

(A-G). Mating duration of *Cs* and *Bully* males paired with a virgin *Cs* female. (A') After mating ended, the proportion of time spent by each male to court performing UWE towards the just mated female was scored for 10 minutes. (B') The proportion of time spent by the males performing UWE an unfamiliar mated female was scored for 10 minutes. (C) Total number of mating reached by *Cs* and *Bully* males over six consecutive mating with virgin *Cs* females. (D) Mating duration of *Cs* and *Bully* males over six consecutive mating with virgin *Cs* females. A significant interaction between males' genotype and number of mating was found, indicating that *Cs* decreased their mating duration with the number of mating, while *Bully* males did not. (E) Proportion of mating that gave rise to progeny in consecutive mating assay. (F') Proportion of time spent by naïve *Cs* and *Bully* males to display UWE either toward *Cs*-mated or *Bully*-mated decapitated female over an observation period of 2 minutes. (G') Proportion of *Cs*-mated and *Bully*-mated females that re-mated 12 or 14 days after a first mating. For graphs A, A', B, B', F, F', G and G', stars indicate significant differences (* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$) and statistical details are given in **Figure 1-table supplement 2**.

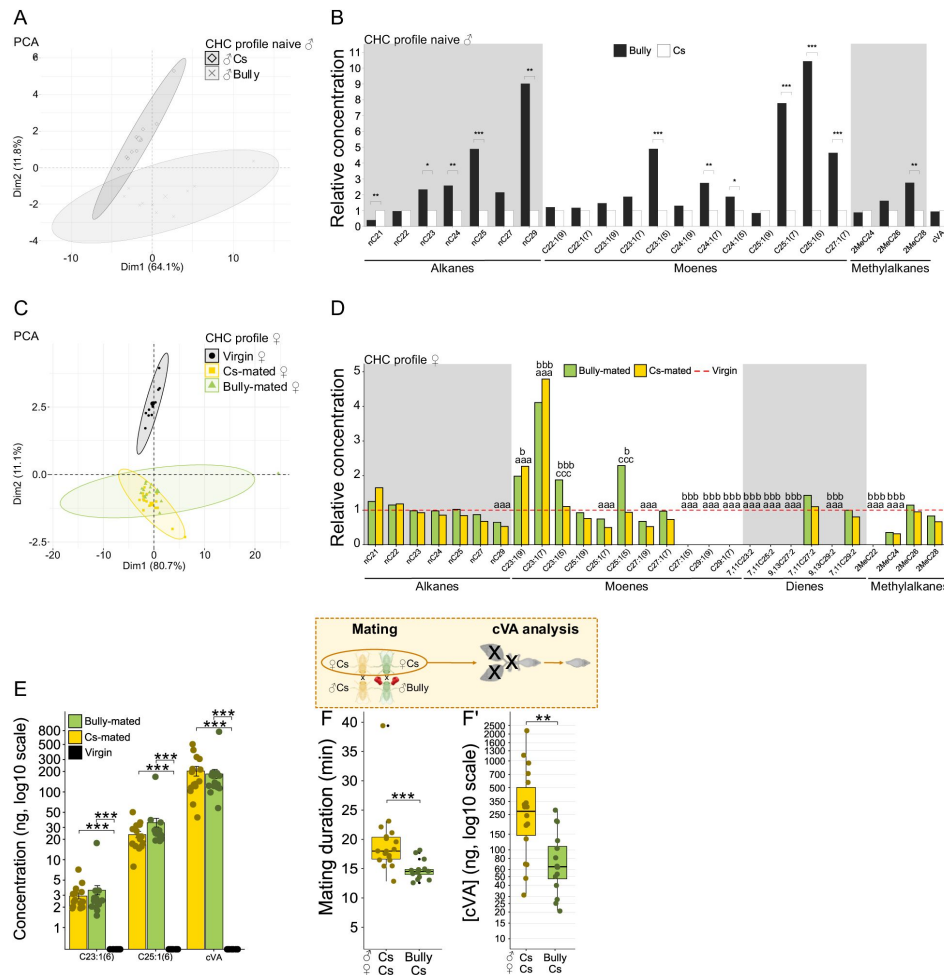


Figure 3.

Hyper-aggressive males exhibit altered CHCs profiles and transfer reduced amounts of mate-guarding pheromones to females.

(A) PCA analysis on mean concentration for each Cuticular Hydrocarbon (CHCs) compound of naïve Cs and *Bully*. Their CHCs profiles were found similar. (B) Relative concentrations of *Bully*'s CHCs were calculated with respect to those of Cs males. The CHCs levels in Cs males were normalized to 1, and *Bully*'s CHC concentrations were expressed relative to this reference. (C) PCA analysis on mean concentration for each CHCs compound of virgin, Cs- and *Bully*-mated females. (D) Relative concentrations of CHCs compounds in mated females compared to virgin females. CHCs levels in virgin females were normalized to 1, and the concentrations in mated females were expressed relative to this baseline. Significant differences are indicated by letters: *a* indicates a significant difference between virgin and Cs-mated females; *b* indicates a significant difference between virgin and *Bully*-mated females; and *c* indicates a significant difference between Cs- and *Bully*-mated females (*a* *P* < 0.05; *aa* *P* < 0.01; *aaa* *P* < 0.001). (E) Mean ± SE concentration in ng of CHCs compounds that were not present in virgin females. (F) Mating duration of Cs and *Bully* males with a Cs female. (F') cVA concentration measured within Cs- and *Bully*-mated genitalia tract right after mating. For graphs E, F and F', stars indicate significant differences (* *P* < 0.05, ** *P* < 0.01, *** *P* < 0.001) and statistical details are given in [Figure 1-table supplement 2](#).

	Mean concentration (ng)						Statistics
	Naive males						♂ Cs vs.
Compounds	♂ Cs (n = 10)			♂ Bully (n = 10)			♂ Bully
Alkanes	252.857	±	33.157	536.232	±	114.016	*
nC21	38.827	±	5.550	15.179	±	2.964	**
nC22	21.293	±	3.344	20.323	±	3.963	
nC23	166.409	±	23.184	387.189	±	81.398	*
nC24	5.767	±	0.814	14.826	±	2.908	**
nC25	17.979	±	2.311	87.979	±	19.520	***
nC27	4.337	±	0.201	3.913	±	1.264	
nC29	0.756	±	0.151	6.823	±	2.593	**
Monoenes	757.996	±	118.420	1927.262	±	393.721	**
C22:1(9)	39.569	±	13.847	47.661	±	26.956	
C22:1(7)	10.415	±	1.736	12.148	±	2.362	
C23:1(9) (9-T)	56.781	±	10.104	82.572	±	15.023	
C23:1(7) (7-T)	473.731	±	87.688	886.163	±	196.863	
C23:1(5) (5-T)	17.895	±	3.644	87.653	±	20.232	***
C24:1(9)	12.995	±	2.508	16.822	±	3.103	
C24:1(7)	3.747	±	0.792	10.267	±	1.434	***
C24:1(5)	2.514	±	0.457	4.696	±	0.670	*
C25:1(9)	63.957	±	9.855	53.126	±	12.710	
C25:1(7) (7-P)	17.615	±	3.102	137.174	±	26.937	***
C25:1(5) (5-P)	54.442	±	10.379	568.837	±	114.334	***
C27:1(7)	4.337	±	0.862	20.144	±	1.264	***
Methyl-alkanes	122.864	±	17.761	233.548	±	54.046	*
2MeC24	5.241	±	1.010	4.541	±	0.678	
2MeC26	82.688	±	12.707	132.912	±	32.207	
2MeC28	34.935	±	5.471	96.095	±	21.9	**
Total CHC	1.914	±	0.261	1.783	±	0.334	

Table 1

Comparison of CHCs concentrations between naïve Cs and Bully males.

Mean ± SE concentrations are provided in ng for each compound and each group of compounds. In statistics columns, stars represent significant differences (* P < 0.05, ** P < 0.01, *** P < 0.001). Statistical details are given in **Figure 1-table supplement 2**.

PCA captures the major part of the variance in the data (**Figure 3—figure supplement 1A**). In addition, some compounds such as 2MeC24, C23:1(6), cVA and C25:1(6) strongly influence PC1 and explain the difference between the clusters, *i.e.* female mated status (**Figure 3—figure supplement 1B-D**). Individual compound analysis revealed that this difference was primarily due to the transfer of male-specific CHCs during mating not found in virgin females (**Figure 3D** and **E**, **Table 2**). Relatively to virgin females, C23:1(9) (9-Tricosene, 9-T), C23:1(7) (7-Tricosene, 7-T), C23:1(6), C25:1(6) and cVA were significantly elevated in mated females (**Figure 3D** and **E**, **Table 2**). These results are consistent with previous studies showing that 7-T and 9-T are acquired by females during mating (Everaerts et al., 2010; Laturney and Billeter, 2016). When comparing the two groups of mated females, individual compound analysis showed that C23:1(5) (5-Tricosene, 5-T) and C25:1(5) (5-Pentacosene, 5-P) were significantly elevated in *Bully*-mated females compared to *Cs*-mated females (**Table 2**, surrounded in red, **Figure 3D**). These findings suggest that 5-P and 5-T might contribute to promoting remating in females that have previously mated with *Bully* males (**Figure 2F** and **G**). Given that *Bully* males also showed higher levels of both 5-P and 5-T compared to naïve *Cs* males (**Figure 3B**), it is likely that the elevated levels of these compounds observed in females result from their transfer during mating.

cVA, a male-specific compound that acts as an anti-aphrodisiac and a mate-guarding molecule, is primarily transferred to the female's genital tract during mating, with only trace amounts deposited on the cuticle (Laturney and Billeter, 2016). Females with higher cVA levels (Kurtovic et al., 2007), combined with 7-T, are less attractive to males (Laturney and Billeter, 2016). Given that remating rate of *Bully*-mated females was higher than that of *Cs*-mated females (**Figure 2G**), we hypothesized that *Bully*-mated females would receive less cVA in their genital tract during mating, thus decreasing their attractiveness than when mating with *Cs* males. To test this, we conducted mating assays with *Cs* females paired with either *Cs* or *Bully* males, followed by dissection of the females' genital tracts to quantify the amount of cVA transferred. We found that females mated with *Bully* males had about three times lower cVA quantities in their genital tracts compared to those mated with *Cs* males (**Figure 3F**). Combined with higher quantities of 5-T and 5-P, reduced cVA level in genital tract likely contribute to the increased attractiveness and higher remating rates observed in females that previously mated with *Bully* males.

Overall, our findings suggest a mechanistic basis for the reduced pre- and post-copulatory behaviors observed in *Bully* males. Differences in CHCs profiles between naïve *Cs* and *Bully* may underlie a reduced attractiveness to females, thereby reducing their mating success. Additionally, increased levels of 5-T and 5-P, combined with reduced cVA transfer to the female genital tract, may contribute to the lower mate-guarding efficiency observed in these hyper-aggressive males.

The lower copulatory strategy of aggressive males is offset by a longer lifespan and maintenance of reproductive success later in life

Based on the main life-history trade-off between reproduction and survival, we hypothesized that the reduced pre- and post-copulatory success in *Bully* males would increase their survival. To test this, we conducted longitudinal courtship assays in which individual *Cs* and *Bully* males were paired weekly with a new virgin *Cs* female until their natural death. Throughout their lifespan, *Bully* males consistently displayed shorter MD than *Cs* males (**Figure 4A**). While *Cs* males increased their MD with age or experience, a pattern reported in another *Drosophila* species (Dhole and Pfennig, 2014), this effect was absent in *Bully* males, whose MD remained stable across mating and throughout their lives (**Figure 4A**). Analysis of individual male slopes confirmed this divergence: most *Cs* males showed increasing MD across mating events, while *Bully* males did not (**Figure 4—figure supplement 1**). Mating success followed a similar pattern. Although *Bully* males mated less frequently than *Cs* males during the first 70 days, they maintained reproductive activity after all *Cs* males had ceased reproductive activity. All *Cs* males

Table 2

Comparison of CHCs concentrations among virgin, Cs-mated, and Bully-mated females.

Mean $\pm$ SE concentrations are provided in ng for each compound and each group of compounds. In the statistics columns, stars indicate significant differences (* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$). Statistical details are given in **Figure 1-table supplement 2**.

Supplement 2

	Mean concentration (ng)						Statistics					
	Virgin Cs ♀			Mated Cs ♀			♂ Cs vs.	♂ Cs vs.	Cs-mated vs.			
Compounds	♂ Cs (n = 10)			Cs-mated ♀ (n = 16)			Bully-mated ♀ (n = 14)			Cs-mated	Bully-mated	Bully-mated
Alkanes	227.22	±	15.64	193.89	±	19.06	220.99	±	44.60			
nC21	4.37	±	0.32	7.20	±	0.74	5.47	±	1.52			
nC22	5.20	±	0.42	6.15	±	0.71	6.02	±	1.49			
nC23	80.02	±	7.20	74.21	±	7.40	78.56	±	11.28			
nC24	10.29	±	0.86	8.84	±	0.98	10.07	±	2.32			
nC25	72.97	±	4.81	61.60	±	5.88	74.89	±	16.93			
nC27	47.91	±	4.39	32.45	±	3.75	41.76	±	10.42			
nC29	6.47	±	0.52	3.44	±	0.30	4.21	±	1.15	***		
Monoenes	171.43	±	11.19	190.32	±	19.14	250.44	±	69.17			
C23:1(9) (9-T)	2.55	±	0.13	5.79	±	0.63	5.07	±	1.16	***	*	
C23:1(7) (7-T)	12.83	±	0.92	61.45	±	6.92	52.76	±	12.45	***	***	
C23:1(6)	0,00			2.91	±	0.09	3.54	±	1.09	***	***	
C23:1(5) (5-T)	2.86	±	0.21	3.16	±	0.35	5.37	±	1.07		*	*
C25:1(9)	18.91	±	1.92	14.40	±	1.74	17.51	±	5.62			
C25:1(7) (7-P)	43.11	±	2.41	21.49	±	2.52	32.11	±	10.52	***		
C25:1(6)	0,00			23.48	±	2.60	35.23	±	10.26	***	***	
C25:1(5) (5-P)	21.85	±	1.43	20.52	±	2.67	50.09	±	14.31		*	***
C27:1(9)	28,00	±	1.94	14.78	±	1.57	18.99	±	4.77	***		
C27:1(7)	30.57	±	2.34	22.35	±	2.77	29.78	±	8.39			
C27:1(5)	7.27	±	5.74	Trace			Trace			***	***	
C29:1(9)	1.56	±	0.12	Trace			Trace			***	***	
C29:1(7)	1.93	±	0.15	Trace			Trace			***	***	
Dienes	319.86	±	28.80	280.70	±	22.38	358.05	±	109.72			
7,11C23:2	4.94	±	0.45	Trace			Trace			***	***	
7,11C25:2	8.33	±	0.77	Trace			Trace			***	***	
9,13C27:2	8.21	±	0.53	Trace			Trace			***	***	
7,11C27:2 (7,11-HD)	151.71	±	17.63	167.05	±	13.62	216.99	±	66.11			
9,13C29:2	5.57	±	0.62	Trace			Trace			***	***	
7,11C29:2	141.11	±	13.85	113.65	±	9.84	141.06	±	43.73			
Methyl-alkanes	226.7	±	23.87	186.84	±	18.92	226.48	±	59.16			
2MeC22	2.61	±	0.17	Trace			Trace			***	***	
2MeC24	16.87	±	1.28	5.33	±	0.67	5.99	±	1.89	***	***	
2MeC26	145.81	±	18.14	138.89	±	14.05	167.05	±	41.77			
2MeC28	64.02	±	6.79	42.63	±	4.58	53.44	±	15.58			
11-cis-vaccenyl acetate (cVA)	0,00			202.66	±	31.99	183.55	±	46.33	***	***	
Total CHC	947.82	±	62.95	1054.41	±	104.31	1239.50	±	327.40			

stopped mating by day 78 and died by day 97, whereas *Bully* males continued to mate until day 98 and survived up to 119 days (**Figure 4B** [↗](#)). Remarkably, these late-life matings produced viable offspring, demonstrating that *Bully* males retain fertility at older ages.

Bully males survived significantly longer than Cs males, suggesting a shift in the reproduction-survival trade-off. To explore this further, Cs virgin females were mated with either Cs and *Bully* males, and progeny survival was monitored. Progeny from *Bully* mating showed a median survival (EC50) of 79 days, with the last male dying at day 106 (**Figure 4C** [↗](#), green), while progeny from Cs males showed an EC50 of 57 days, with the last death occurring at day 90 (**Figure 4C** [↗](#), yellow). This showed that the median survival was extended by ~22 days (79 vs. 57 days) in *Bully* males. We then monitored male survival in mixed-sex groups composed of ten males and twelve females. *Bully* males still showed increased longevity, with an EC50 of 56 days and the last death on day 71, compared to Cs males, that had an EC50 of 43 days and all died by day 67 (**Figure 4D** [↗](#)). These findings are consistent with our previous results (**Figure 4B** [↗](#) and **C** [↗](#)), confirming that *Bully* males have a longer lifespan and can transmit this trait to their offspring.

Our findings demonstrate that males selected for high levels of aggression show reduced investment in mating success but exhibit extended lifespan and sustained reproductive output later in life. Together, these findings support our initial hypothesis that selection for aggressive traits influences a key life-history trade-off, by favoring survival at the expense of reproductive traits.

Discussion

Our study reveals that selection for aggressive traits in *D. mel.* males influences one key life-history trade-off, in which survival is extended while reproductive success is reduced. Our results indicate that selection for aggressive traits directly or indirectly affects both pre- and post-copulatory strategies in *Bully* males, challenging the hypothesis that aggression necessarily enhances mating success (Andersson, 1994 [↗](#); West-Eberhard, 1983 [↗](#)). Aggressive males had lower pre-copulatory success with decreased mating success and shorter mating duration than their less aggressive counterparts. Despite reduced refractory courtship period after mating, *Bully* males fail to achieve higher remating rate and exhibit a less effective mate-guarding strategy, as evidenced by the increased remating rate and potentially attractiveness of their previous mates. Differences between *Bully* and Cs males also extend to chemical profiles, which may account for reduced mating success in *Bully* males. In addition, females mated to highly aggressive males exhibited quantitative differences in CHCs profiles compared those mated with Cs males, along with a dramatic reduction in cVA quantities in their genital tracts after mating, providing a mechanism underlying a less efficient mate-guarding. Our findings suggest that artificial selection applied to male aggressive traits to produce the *Bully* line, is associated with changes in reproductive traits, enhanced survival, and increased CHCs levels, suggesting the existence of pleiotropic genes whose variation links aggression with these other traits.

The increased survival rate observed in highly aggressive males raises questions about the evolutionary consequences of aggression. While *Bully* males exhibit reduced pre- and post-copulatory success compared to Cs males, their extended lifespan allows them to sustain mating activity and produce viable offspring at later ages, challenging the classical view that early reproductive success is the primary driver of male fitness (Flatt, 2011 [↗](#)). This shift suggests that alternative reproductive strategies, even those associated with lower mating success, can persist if they confer long-term survival advantages. Moreover, the extended lifespan of both *Bully* males and their progeny suggests the transmission of pleiotropic genetic traits that influence both aggression and survival, or the transmission of independent genetic loci affecting these traits separately. A previous study identified a limited number of genes that are differentially expressed in the *Bully* line compared to Cs (Chowdhury et al., 2017 [↗](#)). One of these genes, CG13646, a

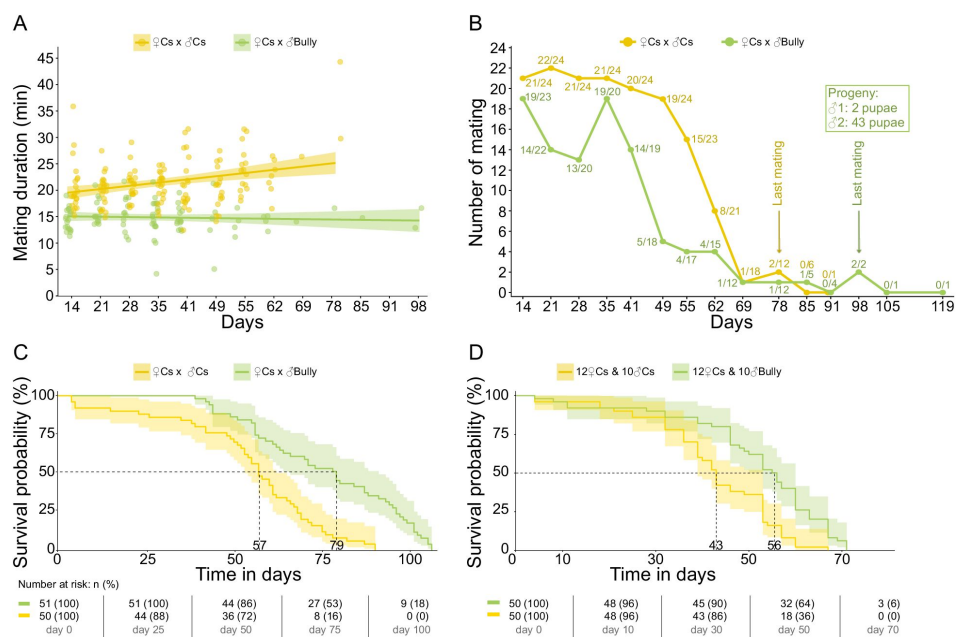


Figure 4.

Aggressive males offset their reduced reproductive success with a longer lifespan and preserved reproductive success in later life.

(A) Mating duration of Cs and Bully males with virgin Cs females across their lifespan. A significant interaction between male genotype and age was observed: Bully males maintained a stable mating duration over time, whereas Cs males showed an increase in mating duration with successive mating events. (B) Number of mating throughout the lifespan of Cs and Bully males. Numbers next to data points indicate the proportion of mating. Arrows denote the last mating event and the time of death for each genotype. The number of offspring produced from the final Bully mating is indicated next to the corresponding arrow. (C) Survival curves of progeny from ♀Cs x ♂Cs and ♀Cs x ♂Bully crosses. Progeny from Bully males had a median survival (EC₅₀) of 79 days, with the last individual dying at day 106 (green line), while progeny from Cs males showed an EC₅₀ of 57 days, with the final death at day 90 (yellow line). (D) Survival curves of males raised in group of 12 ♀Cs & 10 ♂Cs or 12 ♀Cs & 10 ♂Bully. Bully males had a median survival (EC₅₀) of 56 days, with the last individual dying at day 71 (green line), while Cs males showed an EC₅₀ of 43 days, with the final death at day 67 (yellow line). Statistical details are given in [Figure 1-table supplement 2](#).

putative transmembrane transporter, is down-regulated in *Bully* males compared to *Cs*. Future research should investigate the genetic and neurobiological mechanisms underlying the trade-off reproduction/survival by examining the roles of candidate genes whose expression is up- or down-regulated in the *Bully* line.

Despite maintaining high courtship intensity, highly aggressive males exhibit reduced mating success and mating duration, suggesting that aggression is linked to physiological or behavioral traits that may be suboptimal for reproductive performance. For example, males might be less attractive for reasons unrelated to their courtship efforts. Rather than maximizing immediate reproductive output, aggressive males might allocate energy in a way that prioritizes long-term survival, potentially reflecting a life-history shift toward extended reproductive capacity at the cost of reduced mating success.

We showed that naïve highly aggressive individuals present distinct chemical profiles compared to unselected wild-type flies, indicating that selection for aggressive traits may have broader impacts, such as indirectly influencing mate choice through male attractiveness to females. Elevated levels of most CHCs in aggressive males suggest that they could be perceived as more attractive to females than wild-type males. However, despite these chemical traits potentially enhancing female attraction, aggressive males exhibit lower mating success, highlighting that a higher quantity of CHCs alone does not guarantee mating success. However, besides 7-T as CHCs fostering male attractiveness, other cuticular compounds, such as palmitoleic acid signaling through Or47b, modulate courtship behaviors in an age-dependent manner (Dweck et al., 2015 [↗](#); Kohlmeier et al., 2021 [↗](#); Lin et al., 2016 [↗](#)). This mechanism enables older males to initiate courtship more rapidly and with greater intensity, potentially enhancing their mating success relative to younger males (Lin et al., 2016 [↗](#)). In our experiments, we did not measure palmitoleic acid levels in *Bully* and *Cs* males. However, it remains possible that *Bully* males have lower levels of palmitoleic acid compared to *Cs* males. This could explain their decreased mating success by potentially reducing their attractiveness. Aggressive males may signal their fitness through chemical cues, but their high level of aggression could hinder mate choice by females, who may prioritize other factors, such as male courtship behavior or compatibility in post-mating interactions, over chemical signaling alone. This raises the possibility that aggression-driven sexual selection operates through both direct behavioral competition and indirect chemical signaling mechanisms, influencing overall reproductive success.

The influence of aggression extends beyond accessing mates, affecting post-mating behaviors, mate-guarding and chemical communication, shaping male reproductive strategies and female remating dynamics. Hyper-aggressive males exhibit a shortened refractory period to court females showed by an elevated post-mating courtship yet reduced mate-guarding efficiency, likely contributing to increasing remating rate of both males and females respectively. This suggests that while aggression may provide advantages in direct competition, it can incur reproductive costs by diminishing mate-guarding efficiency. The chemical analysis further supports this assumption, showing that females mated with the hyper-aggressive *Bully* males exhibit elevated levels of specific CHCs (5-T and 5-P) and significantly lower cVA transfer into the female genital tract, which may fail to efficiently decrease their attractiveness, thus triggering higher remating rate. This is supported by other study showing that lower cVA levels can shorten the effectiveness of chemical mate-guarding, allowing quicker female re-attractiveness (Ejima, 2015 [↗](#)). As 7-T and cVA act in concert to modulate behavior (Laturney and Billeter, 2016 [↗](#); Verschut et al., 2023 [↗](#)), it is possible that the elevated levels of 5-T and 5-P in *Bully*-mated females interact synergistically too, or with potential other CHCs, to fail to reduce female attractiveness. This would suggest that aggression may indirectly influence post-copulatory sexual selection by changing female chemical signatures.

In summary, our findings emphasize the need to consider aggression within a broader life-history and evolutionary framework. The influence of aggression on the trade-off survival/reproduction, potentially through chemical signaling, illustrates how selection for competitive traits does not

occur in isolation but is shaped by interrelated behavioral and physiological factors. This supports the notion that aggression benefits to individuals in competitive environment but may induce reproductive constraints. By framing aggression as part of a broader life-history strategy rather than an isolated trait, our study highlights the intricate balance between competition, survival, and reproductive success. These insights offer novel perspectives on the evolutionary consequences of aggressive behavior and its role in shaping reproductive strategies in animal populations.

Material and methods

Experimental model and subject details

Fly stocks

Drosophila melanogaster strains were raised at 25°C under a 12h:12h light/dark cycle (LD= 9:00a.m. – 9:00p.m.) on standard fly food (74% corn flour, 28% yeast, 40% sugar, 8% agar, 20% Moldex). Canton-S (Cs) flies (from Edward Kravitz's laboratory, [RRID:BDSC_64349](#)) were used as wild-type. The *Bully* A and B lines were originally selected in Edward Kravitz's laboratory from Cs during 37 generations, resulting in hyper-aggressive male flies ([Garbaczewska et al., 2013](#); [Hopkins et al., 2019](#)).

Experimental arena

Experimental arenas were described in details in [Trannoy et al., 2015](#) ([Trannoy et al., 2015](#)). Briefly, resin blocks containing three circular behavioral arenas (dimension: 2.3 cm diameter, 1.7 cm height) were used to conduct behavioral assays. Each arena was divided in two equal sizes by a plastic divider allowing flies to acclimate without physical interactions with the second fly. Then, dividers are removed from the arenas to start the behavioral assays. All behavioral assays were performed in empty arenas, except for aggression assays and courtship assays with decapitated females, in which a food cup (1.5 cm diameter, 1 cm height) containing fresh standard fly food with a drop of fresh yeast paste on the surface was placed in the center of each arena.

Method details

Commonalities between all behavioral assays

Late-stage pupae were collected from stocks raised at 25°C. Male pupae were individually placed in 5ml vials (Dutscher, ref 390597) containing about 1ml of standard fly food while female pupae were placed in group of 15 in ≈47ml vials (Dutscher, ref 789001B) containing about 15ml of standard fly food. All vials were placed at 25°C for 7 days. Seven-day-old individuals were randomly inserted in arenas by negative geotaxis, and allow to acclimate for 5 min before experiments. All behavioral assays were performed at 25°C. All tests were recorded with Basler camera Ace1970 through the StreamPix8 multi video recording software ([norpix.com](#)) and analyzed with BORIS software ([Friard and Gamba, 2016](#)). After 15 min or 30 min, if pairs of flies had not yet lunged or mated respectively, we considered that no aggression or mating occurred and arenas were not considered for analysis. No power analyses were performed prior to the experiments, as we chose to use sample sizes of approximately 20 replicates, which is considered standard practice in the field.

Aggression assays

Two days prior to the aggression assays, males were anesthetized with CO₂, and half were marked with a small white dot on the thorax before being returned to their original isolation vials. Aggression tests were conducted between ZT0 and ZT3 using size-matched male pairs. Males competed for access to a central food cup containing yeast. The following behaviors were recorded for 10 minutes following the first lunge: latency to lunge (i.e., the time between the initial encounter on the food cup and the first lunge), total number of lunges, and number of boxing events.

Mating assays

Seven-day-old individuals were tested in arenas without any food source. Latency to court and to mate, defined as the time between the first encounter and either the first Unilateral Wing Extension (UWE) or the initiation of mating, respectively, were measured. Total UWE duration was recorded over a 10-minute period starting from the first encounter. Mating proportion and mating duration were also scored. In all assays, males were reared in social isolation, except for **Figure 1H** [↗](#), where they were housed in groups of 10 males for 7 days.

Mating assays with decapitated females

Females were anesthetized with CO₂ and placed under the binocular for head removal. Experiments with decapitated females were performed in arenas with food cup filled with fresh fly food but without yeast.

Multiple consecutive mating events assays

Single naïve Cs and *Bully* males were sequentially paired with six virgin Cs females. For each male, mating duration and the number of successful mating were recorded. After each copulation, the mated female was transferred to a 5 ml vial containing fly food to assess progeny production.

Mating assays throughout the lifespan

24 Cs and *Bully* males were reared in isolation and paired with seven-day-old virgin Cs females for mating assays, conducted every seven days until all males had died. At each time point, mating duration and mating success (proportion of mating) were recorded. After all Cs males had died, females that had mated with *Bully* males were transferred to 5 ml vials containing fly food to assess progeny production.

Attractiveness assays

Measure of post-mating UWE

Mating assays were performed by pairing Cs and *Bully* males with Cs females. Right after mating end, we scored the total time that males displayed UWE for 10 minutes towards the just-mated females.

Measure of post-mating UWE with switched females

Mating assays were performed with naïve Cs or *Bully* males and virgin Cs females. Immediately after mating ends, females were switched: Cs males interacted with *Bully*-mated females and *Bully* males interacted with Cs-mated females. Then total time of UWE towards the females was scored during 10 min.

Measure of post-mating UWE with decapitated females

Mating assays were conducted using naïve *Cs* or *Bully* males paired with virgin *Cs* females. Following copulation, mated females were decapitated. One decapitated *Cs*-mated and one *Bully*-mated female were then placed on the surface of the food cup. A new naïve *Cs* or *Bully* male was introduced and allowed to interact with both decapitated females. The duration of Unilateral Wing Extensions (UWE) directed toward each female was recorded over a 2-minute period following the first encounter. Arenas in which males interacted with only one of the two females were excluded from the analysis, as no choice could be determined.

Re-mating assays

Mating assays were performed with naïve *Cs* or *Bully* males and virgin *Cs* females. Mated-females were then individually placed in 5ml vials containing about 1ml of standard fly food and kept at 25°C. 12 or 14 days after the initial mating, mated-females were paired with a new naïve *Cs* male in empty arena and the proportion of re-mating during 30 min was scored.

Survival assays

Survival with isolated males

Late-stage pupae from the progeny of $\varnothing Cs \times \sigma Cs$ and $\varnothing Cs \times \sigma Bully$ crosses were individually isolated in 5 ml vials containing 1ml of standard fly food and kept at 25 °C until adult emergence. The number of dead individuals was recorded daily until all flies had died. To maintain food quality, flies were transferred weekly to fresh vials containing approximately 1 ml of new fly food.

Survival with group-raised males

Late-stage *Cs* and *Bully* male pupae and *Cs* female pupae were placed in group of 12 $\varnothing Cs$ -10 σCs or 12 $\varnothing Cs$ -10 $\sigma Bully$ in ≈ 47 ml vials containing 15mL standard fly food and kept at 25 °C until adult emergence. The number of dead males was recorded daily until they all had died. To maintain food quality, flies were transferred twice a week to fresh vials containing approximately 15 mL of new fly food.

CHCs analysis

Extraction and analysis of cuticular hydrocarbons (CHCs) of naïve *Cs* and *Bully* males, as well as virgin *Cs*, *Cs*- and *Bully*-mated females were performed in Jean-Christophe Billeter's lab, Groningen, Netherlands, with a Gas Chromatography coupled with Flame Ionization Detection (GC-FID) as already described (Laturney and Billeter, 2016 [\[24\]](#)). Briefly, single whole flies were put in a 2ml glass vial (Supelco certified vial kit, 2ml clear glass vial 12×32mm and Supelco inserts) containing 50µl of nC18+nC26 10 ng/µl standard solution (made in Billeter's lab) diluted in n-Hexane 99%+ (Acrös Organic®). Vials were then vortexed at the lowest speed for 2min, afterward flies' body were gently removed from vials using a "J-shape" paper clip sliced at the end, taking care not to damage the body to prevent from hemolymph contamination. Finally, samples were placed in GC-FID for apolar compounds analysis only.

cVA quantification into genital tract

Mating assays were performed with naïve *Cs* or *Bully* males and virgin *Cs* female. Right after mating, females were transferred into a 5ml vial previously put in ice in -80°C fridge to make them fall asleep quickly and prevent from sperm ejection. Mated-females' genital tract was dissected in PBS with precision forceps under a binocular loupe. Protocol was already employed in (Verschut et al., 2022 [\[25\]](#)). Briefly, abdomen was torn to make ovaries and genital tract out and visible, then ovaries, oviduct, spermathecae, parovaria and cuticula were removed to keep only the uterus,

taking care not to injure it to prevent from sperm leak. Each uterus was inserted in a 2ml glass vial (Supelco certified vial kit, 2ml clear glass vial 12×32mm and Supelco inserts) containing 50µl of n-Hexane 99%+ (Acrös Organic) for 1h at ambient temperature. Hexane solution was then evaporated under helium flow to prevent from cVA oxidation with dioxygen. Samples were sent to Jean-Christophe Billeter's lab for GC-FID analysis of cVA, as describe above in CHCs analysis method part.

Quantification and statistical analysis

No data have been removed from the statistical analysis. Some experiments were independently replicated by different authors to reduce potential bias. All statistical analyses were performed on R software version 4.0. (Team, 2020 [\[link\]](#)) with packages (Fox and Weisberg, 2019 [\[link\]](#)), coin (Hothorn et al., 2006 [\[link\]](#)), dplyr (Wickham et al., 2023 [\[link\]](#)), factoextra (Kassambara and Mundt, 2020 [\[link\]](#)), FactoMineR (Lê et al., 2008 [\[link\]](#)), lme4 (Bates et al., 2015 [\[link\]](#)), lmerTest (Kuznetsova et al., 2017 [\[link\]](#)), MASS (Venables and Ripley, 2002 [\[link\]](#)), MuMIn (Bartoń, 2023 [\[link\]](#)), nlme (Pinheiro et al., 2022 [\[link\]](#)), performance (Lüdtke et al., 2021 [\[link\]](#)), pgirmess (Giraudeau. P, 2018 [\[link\]](#)), predictmeans (Luo et al., 2021 [\[link\]](#)), survival (Therneau, 2022 [\[link\]](#)), vegan (Oksanen J, 2022).

Comparisons of means ($k = 2$) were performed with a T-test when possible (*i.e.* mating duration) otherwise a Fisher Pitman permutation test was used (*i.e.* UWE, latency to court and to mate). For comparisons with more than 2 conditions, a linear model (LM) was used and tested either with a type II ANOVA when possible (*i.e.* mating duration) or a permutation test instead (*i.e.* latency to lunge and CHC concentrations). For comparison of proportions and counting events, a type II analysis of deviance table with a Likelihood Ratio Test (*i.e.* mating success, proportion of female re-mating, number of lunge and boxing events) or a Wald Chi-square test (*i.e.* number of consecutive mating events) was performed. A Principal Component Analysis (PCA) was performed to see different CHC profiles and tested with a PerMANOVA and a SIMPER analysis. Survival data were analysed with a Weibull regression model and a Likelihood Ratio Test.

When paired data, dependency was considered thanks to linear mixed model (LMM) with the source of dependency as random factor. Data following a distribution other than a normal distribution (*e.g.* binomial or Poisson distribution) were computed in a generalized linear model (GLM) (*i.e.* mating success, number of consecutive mating events, number of mating giving progeny, proportion of females re-mating, number of lunges and boxing events). When necessary, interactions were considered within models and tested. All post-hoc comparisons were performed with a sequential Bonferroni correction on the alpha level using either Holm method (Holm, 1979 [\[link\]](#)) when few numbers of comparisons or Benjamini and Hochberg method (Benjamini and Hochberg, 1995 [\[link\]](#)) when more comparisons. All statistical details are given in Table S1.

Acknowledgements

We thank the members of EXPLAIN and IVEP team at the CRCA for their helpful discussion and support on this study. We thank Karvitz's lab for sharing the *Bully* lines. We also thank all the member of Billeter's lab to host A. Defert and train him to GC-FID methodology. We acknowledge support from the French Research National Agency (ANR) (ANR-19-CE37-0018-01 to S.T.), the Fondation Fyssen (190573 to S.T.), and the Doctoral Mobility Campaign of Toulouse III University in 2023 (to A.D.).

Funders had no role in study design, data collection and analysis, decision to publish or preparation of the manuscript.

Additional information

Resource availability

Further information and requests for resources and reagents are accessible from the lead contact, Séverine Trannoy (severine.trannoy@univ-tlse3.fr).

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Conceptualization: S.T., J.C.B., A.D.

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Project administration: S.T.

Supervision: S.T., J.C.B.

Writing: S.T., J.C.B., A.D.

Supporting informations

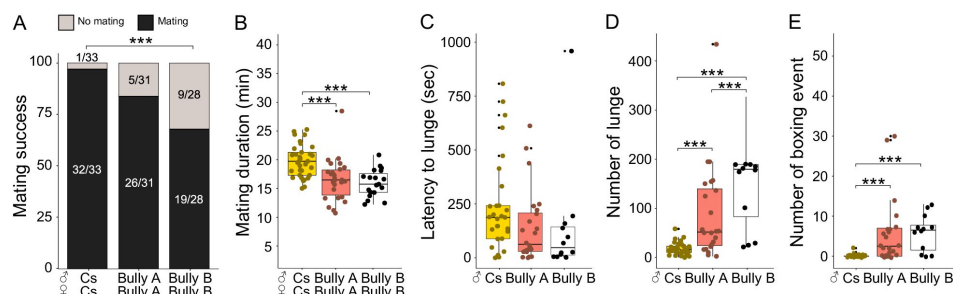


Figure 1-figure supplement 1.

Courtship and aggressive behaviors in *Cs*, *Bully A*, and *Bully B* lines.

(A) Non-competitive mating success of *Cs*, *Bully A* and *Bully B* males paired with a virgin *Cs* female. (B) Mating duration of *Cs*, *Bully A* and *Bully B* males paired with a virgin *Cs* female. (C) Latency to lunge of *Cs*, *Bully A* and *Bully B* males in aggression assay with a territory against a male of the same line. (D) Number of lunges and (E) boxing events of *Cs*, *Bully A* and *Bully B* male pairs in aggression assays. For all graphs, stars indicate significant differences (* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$). Statistical details are given in [Figure 1-table supplement 2](#).

Figure	Statistic model	Results
Figure1		
A	glm(mating~male_genotype, family=binomial(link="logit"), data)	A generalize linear model (GLM) for binomial data and a type II analysis of deviance table (Likelihood Ratio Test, $P=0.329$, $LR\chi^2=0.951$) were performed to compare Cs and Bully males mating success.
B	glm(mating~male_genotype, family=binomial(link="logit"), data)	A generalize linear model (GLM) for binomial data and a type II analysis of deviance table (Likelihood Ratio Test, $P=0.037$, $LR\chi^2=4.347$) were performed to compare Cs and Bully males mating success.
C		A Fisher Pitman permutation test ($P=0.877$, $t=-0.131$) was performed to compare the proportion of time spent by Cs and Bully males to court the virgin female.
D		A Fisher Pitman permutation test ($P=0.319$, $Z=1.043$) was performed to compare Cs and Bully males' latency to court.
E		A Fisher Pitman permutation test ($P=0.266$, $Z=1.170$) was performed to compare Cs and Bully males' latency to mate.
F		A T-test ($P=6.009 \times 10^{-10}$, $t=-6.917$) was performed to compare Cs and Bully males' mating duration.
G		A T-test ($P=0.022$, $t=-2.367$) was performed to compare Cs and Bully males' mating duration.
H	lm(mating_duration~condition, data)	A linear model (LM) and a type II ANOVA (F-test, $P=7.582 \times 10^{-11}$, $F=21.669$) were performed to compare Cs and Bully males' mating duration among raising condition. A post-hoc analysis was performed with a sequential Bonferroni correction after Holm on the alpha level: P_{Cs} .

Figure 1-table supplement 2.

Statistical details of all analyses.

This table gathers all statistical analyses performed in all the Figures. The first column refers to the panels of each Figure, the second one gives the statistical model used to analyze data and the last one provides the names of statistical tests and corrections performed as well as all p-values and statistical values. Statistical details are given in **Figure 1-table supplement 2**.

		isolated_Cs-group=0.013, F=6.583, $P_{\text{Bully-isolated_Bully-group}}=2.532 \times 10^{-4}$, F=15.731, $P_{\text{Cs-isolated_Bully-isolated}}=4.263 \times 10^{-6}$, F=27.697, $P_{\text{Cs-group_Bully-group}}=1.884 \times 10^{-4}$, F=16.032.
Figure 2		
A		A T-test ($P=2.912 \times 10^{-11}$, $t=-8.759$) was performed to compare Cs and Bully males' mating duration.
A'		A Fisher Pitman permutation test ($P<0.001$, $Z=3.239$) was performed to compare Cs and Bully males' time spent to court the mated female.
B		A T-test ($P=5.425 \times 10^{-7}$, $t=-5.818$) was performed to compare Cs and Bully males' mating duration.
B'		A Fisher Pitman permutation test ($P=0.007$, $Z=-2.679$) was performed to compare Cs and Bully males' time spent to court the mated female.
C	glmer(mating~male_genotype + (1 male_id), family = binomial(link = "logit"), data)	A type II analysis of deviance table (Wald Chi-square test, $P=0.659$, $\chi^2=0.195$) was performed to compare Cs and Bully males' total number of mating.
D	lmer(mating_duration~couples*mating + (1 male_id), data)	A linear mixed model (LMM) adjusted by REML and a Satterthwaite T-test were performed to test the effect of males' genotype, number of mating and their interaction on mating duration, considering paired data with males' identity as random factor ($P_{\text{interaction}}=0.001$, $t=-3.261$).
E	glmer(progeny~male_genotype + mating + (1 male_id), family=binomial(link="logit"), data)	A generalized linear mixed model (GLMM) and a Wald Chi-square test were performed to test the effect of males' genotype and number of mating on number of mating giving progeny, considering paired data with males' identity as random factor ($P_{\text{male_genotype}}=0.279$, $\chi^2=1.173$, $P_{\text{mating}}=0.847$, $\chi^2=2.019$).
F		A T-test ($P=7.139 \times 10^{-4}$, $t=-3.617$) was performed to compare Cs and Bully males' mating duration.
F'	lme(UWE~male_genotype + mated_female, random=~1 male_id, data)	A linear mixed effect model and a permutation test (Monte-Carlo test, $P_{\text{male_genotype}}<0.001$, $P_{\text{mated_female}}=0.457$) was performed to compare Cs and Bully males' time spent to court the mated female and if Cs- and Bully-mated females received the same amount of courtship.
G		A T-test ($P=8.743 \times 10^{-5}$, $t=-4.255$) was performed to compare Cs and Bully males' mating duration.
G'	glm(remating~mated_female + day, family = binomial(link = "logit"), data)	A generalize linear model (GLM) for binomial data and a type II analysis of deviance table (Likelihood Ratio Test, $P_{\text{mated_female}}=3.639 \times 10^{-4}$, $LR\chi^2=12.708$, $P_{\text{day}}=0.385$, $LR\chi^2=0.756$) were performed to compare Cs- and Bully-mated re-mating proportion.
Figure 3		
A		A Principal Component Analysis (PCA) and a PerMANOVA were performed to test the effect of males' genotype on

Figure 1-table supplement 2. (continued)

		Euclidian distances of individuals on CHC compounds (P=0.002, F=6.789, R ² =0.274).
B + Table 1	lme(concentration~male_g enotype, random=~1 male_id, data)	A linear mixed effect model and a permutation test (Monte-Carlo test, P=0.009) were performed to compare CHC concentrations between naïve Cs and bully males. A post-hoc analysis was performed with a sequential Bonferroni correction after Benjamini and Hochberg on the alpha level: P _{Alkanes} =0.036, P _{Monoenes} =0.002, P _{Methyl alkanes} =0.048, P _{nC21} <0.001, P _{C22:1(9)} =0.892, P _{C22:1(7)} =0.568, P _{cVA} =0.731, P _{nC22} =0.854, P _{C23:1(9)} =0.175, P _{C23:1(7)} =0.062, P _{C23:1(5)} <0.001, P _{nC23} =0.014, P _{C24:1(9)} =0.345, P _{C24:1(7)} <0.001, P _{C24:1(5)} =0.015, P _{nC24} =0.002, P _{2MeC24} =0.548, P _{C25:1(9)} =0.524, P _{C25:1(7)} <0.001, P _{C25:1(5)} <0.001, P _{nC25} <0.001, P _{2MeC26} =0.159, P _{C27:1(7)} <0.001, P _{nC27} =0.078, P _{2MeC28} =0.003, P _{nC29} =0.005.
C		A Principal Component Analysis (PCA) and a PerMANOVA were performed to test the effect of females' mating status on Euclidian distances of individuals on CHC compounds (P=0.017, F=2.594, R ² =0.123). A post-hoc analysis was performed with a sequential Bonferroni correction after holm on the alpha level: P _{virgin_Cs-mated} =0.001, F=19.434, R ² =0.447, P _{virgin_Bully-mated} =0.034, F=2.266, R ² =0.093, P _{Cs-mated_Bully-mated} =0.746, F=0.428, R ² =0.015.
D + E + Table 2	lme(concentration~female _status, random=~1 female_id, data)	A linear mixed effect model and a permutation test (Monte-Carlo test, P=0.618) were performed to compare CHC concentrations between naïve Cs and bully males. Even though the p-value was not significant, a post-hoc analysis was performed with a sequential Bonferroni correction after Benjamini and Hochberg on the alpha level: P _{Alkanes} =0.705, P _{Monoenes} =0.294, P _{Dienes} =0.715, P _{Methyl alkanes} =0.750, P _{nC21} =0.158, P _{nC22} =0.927, P _{7,11C23:2} <0.001, P _{2MeC22} <0.001, P _{C23:1(9)} =0.023, P _{C23:1(7)} =0.001, P _{C23:1(6)} <0.001, P _{C23:1(5)} =0.016, P _{nC23} =0.908, P _{nC24} =0.899, P _{7,11C25:2} <0.001, P _{2MeC24} <0.001, P _{C25:1(9)} =0.832, P _{C25:1(7)} =0.047, P _{C25:1(6)} <0.001, P _{C25:1(5)} <0.001, P _{nC25} =0.769, P _{9,13C27:2} <0.001, P _{7,11C27:2} =0.747, P _{2MeC26} =0.884, P _{C27:1(9)} =0.016, P _{C27:1(7)} =0.625, P _{C27:1(5)} <0.001, P _{nC27} =0.367, P _{9,13C29:2} <0.001, P _{7,11C23:2} =0.899, P _{2MeC28} =0.413, P _{C29:1(9)} <0.001, P _{C29:1(7)} <0.001, P _{nC29} =0.022, P _{cVA} =0.001. A second post-hoc analysis was performed with significant p-values applying a sequential Bonferroni correction after Holm on the alpha level: P _{7,11C23:2 virgin_Cs-mated} <0.001, P _{7,11C23:2 virgin_Bully-mated} <0.001, P _{7,11C23:2 Cs-mated_Bully-mated} =1, P _{2MeC22 virgin_Cs-mated} <0.001, P _{2MeC22 virgin_Bully-mated} <0.001, P _{2MeC22 Cs-mated_Bully-mated} =1, P _{C23:1(9) virgin_Cs-mated} <0.001, P _{C23:1(9) virgin_Bully-mated} =0.021, P _{C23:1(9) Cs-mated_Bully-mated} =0.641, P _{C23:1(7) virgin_Cs-mated} <0.001, P _{C23:1(7) virgin_Bully-mated} <0.001, P _{C23:1(7) Cs-mated_Bully-mated} =0.583, P _{C23:1(6) virgin_Cs-mated} <0.001, P _{C23:1(6) virgin_Bully-mated} <0.001, P _{C23:1(6) Cs-mated_Bully-mated} =0.803, P _{C23:1(5) virgin_Cs-}

Figure 1-table supplement 2. (continued)

		mated=0.535, $P_{C23:1(5)} \text{ virgin_Bully-mated}=0.019$, $P_{C23:1(5)} \text{ Cs-mated_Bully-mated}=0.020$, $P_{7,11C25:2} \text{ virgin_Cs-mated}<0.001$, $P_{7,11C25:2} \text{ virgin_Bully-mated}<0.001$, $P_{7,11C25:2} \text{ Cs-mated_Bully-mated}=1$, $P_{2MeC24} \text{ virgin_Cs-mated}<0.001$, $P_{2MeC24} \text{ virgin_Bully-mated}<0.001$, $P_{2MeC24} \text{ Cs-mated_Bully-mated}=0.889$, $P_{C25:1(7)} \text{ virgin_Cs-mated}<0.001$, $P_{C25:1(7)} \text{ virgin_Bully-mated}=0.514$, $P_{C25:1(7)} \text{ Cs-mated_Bully-mated}=0.402$, $P_{C25:1(6)} \text{ virgin_Cs-mated}<0.001$, $P_{C25:1(6)} \text{ virgin_Bully-mated}<0.001$, $P_{C25:1(6)} \text{ Cs-mated_Bully-mated}=0.246$, $P_{C25:1(5)} \text{ virgin_Cs-mated}=0.746$, $P_{C25:1(5)} \text{ virgin_Bully-mated}=0.021$, $P_{C25:1(5)} \text{ Cs-mated_Bully-mated}<0.001$, $P_{9,13C27:2} \text{ virgin_Cs-mated}<0.001$, $P_{9,13C27:2} \text{ virgin_Bully-mated}<0.001$, $P_{9,13C27:2} \text{ Cs-mated_Bully-mated}=1$, $P_{C27:1(9)} \text{ virgin_Cs-mated}<0.001$, $P_{C27:1(9)} \text{ virgin_Bully-mated}=0.114$, $P_{C27:1(9)} \text{ Cs-mated_Bully-mated}=0.521$, $P_{C27:1(5)} \text{ virgin_Cs-mated}<0.001$, $P_{C27:1(5)} \text{ virgin_Bully-mated}<0.001$, $P_{C27:1(5)} \text{ Cs-mated_Bully-mated}=1$, $P_{9,13C27:2} \text{ virgin_Cs-mated}<0.001$, $P_{9,13C27:2} \text{ virgin_Bully-mated}<0.001$, $P_{9,13C27:2} \text{ Cs-mated_Bully-mated}=1$, $P_{C29:1(9)} \text{ virgin_Cs-mated}<0.001$, $P_{C29:1(9)} \text{ virgin_Bully-mated}<0.001$, $P_{C29:1(9)} \text{ Cs-mated_Bully-mated}=1$, $P_{C29:1(7)} \text{ virgin_Cs-mated}<0.001$, $P_{C29:1(7)} \text{ virgin_Bully-mated}<0.001$, $P_{C29:1(7)} \text{ Cs-mated_Bully-mated}=1$, $P_{nC29} \text{ virgin_Cs-mated}<0.001$, $P_{nC29} \text{ virgin_Bully-mated}=0.119$, $P_{nC29} \text{ Cs-mated_Bully-mated}=0.781$, $P_{cVA} \text{ virgin_Cs-mated}<0.001$, $P_{cVA} \text{ virgin_Bully-mated}<0.001$, $P_{cVA} \text{ Cs-mated_Bully-mated}=0.754$.
F		A T-test ($P=0.003$, $t=-3.359$) was performed to compare Cs and Bully males' mating duration.
F'		A Fisher Pitman permutation test ($P=0.001$, $Z=-2.390$) was performed to compare cVA concentration received by Cs- and Bully-mated females.
Figure 4		
A	<code>lmer(mating_duration~male_genotype*day + (1 male_id), data, control = lmerControl(optimizer = "Nelder_Mead"))</code>	A linear mixed model (LMM) adjusted by REML and optimized with the Nelder Mead method was performed with a Satterthwaite T-test to test the effect of males' genotype, day and their interaction on mating duration, considering paired data with males' identity as random factor ($P_{\text{interaction}}=3.360 \times 10^{-5}$, $t=4.227$).
B	<code>glmer(mating~male_genotype*day + (1 male_id), family = binomial(link = "logit"), data)</code>	A generalize linear mixed model (GLM) for binomial data fitted by maximum likelihood estimation (MLE) using the Laplace approximation was performed with a Wald Z-test to assess the effect of males' genotype, day and their interaction on number of mating ($P_{\text{interaction}}=0.039$, $Z=-2.068$).
C	<code>survreg(Surv(day, death)~male_genotype, dist="weibull", data)</code>	A survival analysis was performed using a Weibull regression model. The model was globally significant (Likelihood Ratio Test, $\chi^2=19.18$, $p=2.5 \times 10^{-4}$), indicating that parent's genotype had a significant effect on males' survival (Wald test: $\beta=0.287$, $p=6 \times 10^{-6}$, $Z=4.7$).
D	<code>coxph(Surv(day, death)~male_genotype + strata(group), data)</code>	A Cox proportional hazards regression model with stratification by group was performed. The model was globally significant (Likelihood Ratio Test, $\chi^2=15.94$,

Figure 1-table supplement 2. (continued)

		p=7x10 ⁻⁵), indicating that genotype had a significant effect on males' survival (Wald test: β = -0.938, p=8.81x10 ⁻⁵ , Z= -3.921).
Figure 1-figure supplement 1		
A	glm(mating~couples, family=binomial(link="logit"), data)	A generalize linear model (GLM) for binomial data and a type II analysis of deviance table (Likelihood Ratio Test, P=0.005, LR χ^2 =10.596) were performed to compare Cs, Bully A and Bully B males mating success. A post-hoc analysis was performed with a sequential Bonferroni correction after holm on the alpha level: P _{Cs_BullyA} =0.058, LR χ^2 =3.605, P _{Cs_BullyB} =0.001, LR χ^2 =10.596, P _{BullyA_BullyB} =0.147, LR χ^2 =2.100.
B	lm(mating_duration~male_genotype, data)	A generalize linear model (GLM) for binomial data and a type II analysis of deviance table (Likelihood Ratio Test, P=0.005, LR χ^2 =10.596) were performed to compare Cs, Bully A and Bully B males mating success. A post-hoc analysis was performed with a sequential Bonferroni correction after holm on the alpha level: P _{Cs_BullyA} =0.058, LR χ^2 =3.605, P _{Cs_BullyB} =0.001, LR χ^2 =10.596, P _{BullyA_BullyB} =0.147, LR χ^2 =2.100.
C	lm(latency_lunge~male_genotype, data)	A linear model and a permutation test (Monte-Carlo test, P=0.222) was performed to compare Cs, Bully A and Bully B males' latency to lunge.
D	glm(lunge~male_genotype, family=poisson(link="log"), data)	A generalize linear model (GLM) for binomial data and a type II analysis of deviance table (Likelihood Ratio Test, P=2.2x10 ⁻¹⁶ , LR χ^2 =2501.8) were performed to compare Cs, Bully A and Bully B males' number of lunges. A post-hoc analysis was performed with a sequential Bonferroni correction after holm on the alpha level: P _{Cs_BullyA} =2.2x10 ⁻¹⁶ , LR χ^2 =1340.1, P _{Cs_BullyB} =2.2x10 ⁻¹⁶ , LR χ^2 =2293, P _{BullyA_BullyB} =2.2x10 ⁻¹⁶ , LR χ^2 =268.65.
E	glm(boxing~male_genotype, family=poisson(link="log"), data)	A generalize linear model (GLM) for binomial data and a type II analysis of deviance table (Likelihood Ratio Test, P=2.2x10 ⁻¹⁶ , LR χ^2 =213.52) were performed to compare Cs, Bully A and Bully B males' number of boxing events. A post-hoc analysis was performed with a sequential Bonferroni correction after holm on the alpha level: P _{Cs_BullyA} =2.2x10 ⁻¹⁶ , LR χ^2 =185.54, P _{Cs_BullyB} =2.2x10 ⁻¹⁶ , LR χ^2 =149.06, P _{BullyA_BullyB} =0.729, LR χ^2 =0.120.
Figure 3-figure supplement 1		
A		PC1 and PC2 eigenvectors for CHC profiles analysis.
B, C, D		A SIMPER analysis was performed on Bray-Curtis distance matrix (CHC compounds) and tested with a permutation test. Due to a huge number of comparisons, only significant p-values are given here: • Virgin ♀ vs. Cs-mated ♀ : P_{cVA}<0.001, P_{C23:1(7)}<0.001, P_{2MeC28}=0.023, P_{C25:1(7)}<0.001, P_{C25:1(6)}<0.001, P_{nC27}=0.008,

Figure 1-table supplement 2. (continued)

		$P_{C27:1(9)} < 0.001$, $P_{2MeC24} < 0.001$, $P_{nC29} < 0.001$, $P_{C23:1(9)} < 0.001$, $P_{C23:1(6)} < 0.001$ • Virgin ♀ vs. Bully-mated ♀ : $P_{cVA} = 0.006$, $P_{7,11C29:2} = 0.007$, $P_{2MeC28} = 0.008$, $P_{C25:1(6)} < 0.001$, $P_{C25:1(7)} < 0.001$, $P_{nC27} = 0.006$, $P_{C27:1(9)} < 0.001$, $P_{2MeC24} < 0.001$, $P_{C25:1(9)} = 0.006$, $P_{nC29} < 0.001$, $P_{C23:1(6)} < 0.001$, $P_{C23:1(6)} < 0.001$ • Cs-mated ♀ vs. Bully-mated ♀ : $P_{C25:1(5)} < 0.001$, $P_{nC21} < 0.001$, $P_{C23:1(5)} = 0.003$.
Figure 4-figure supplement 1		
		Interindividual variability across multiple mating events throughout Cs and Bully males' lifespan.

Figure 1-table supplement 2. (continued)

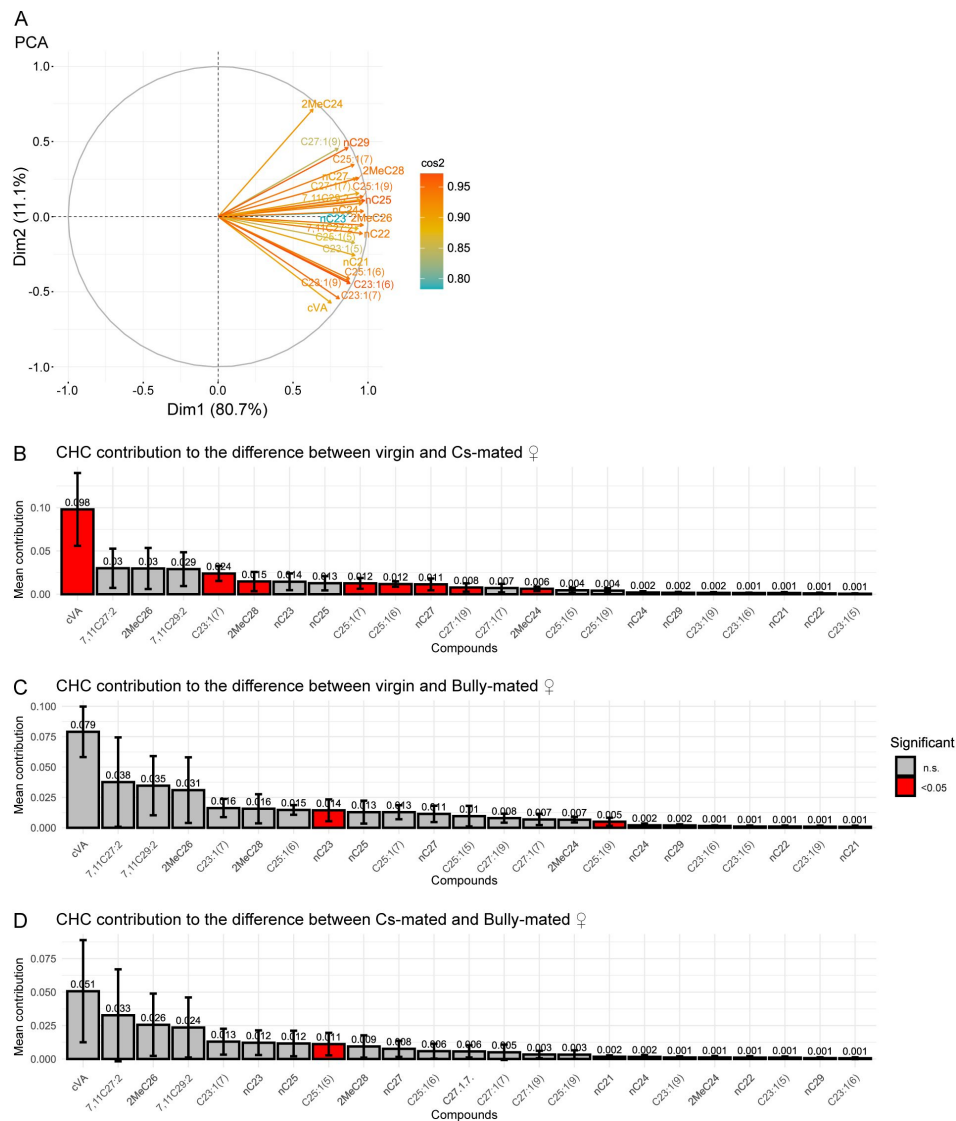


Figure 3-figure supplement 1.

More details on PCA analysis for virgin and mated females.

(A) Eigenvectors and associated $\cos^2$ for each CHCs compounds. Contribution of each CHCs compound in the difference between (B) virgin and Cs-mated, (C) virgin and *Bully*-mated and (D) Cs-mated and *Bully*-mated CHCs profiles. Compounds contributing significantly to the difference between CHCs profiles are represented in red. Numbers above bars refer to the mean contribution of each compound in the difference between CHCs profiles. Statistical details are given in **Figure 1-table supplement 2**.

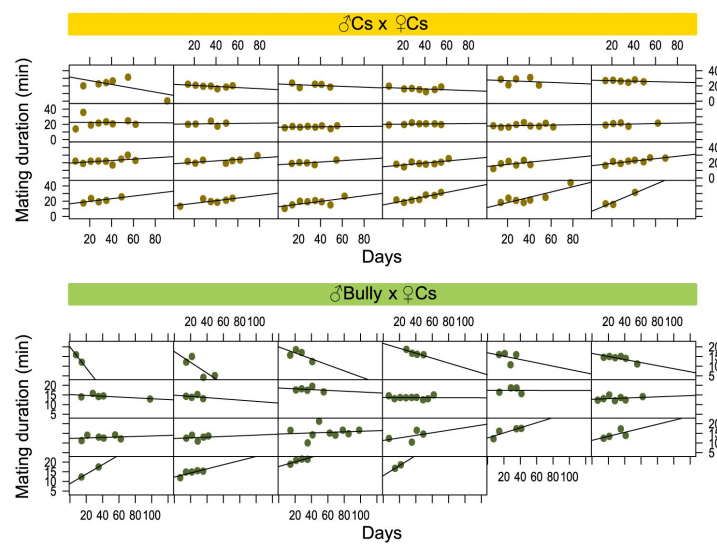


Figure 4-figure supplement 1.

Interindividual variability across multiple mating events throughout the flies' lifespan.

Interindividual variability in mating duration for each (A) *Cs* and (B) *Bully* male when mated with a virgin *Cs* female across their lifespan. Each panel represents the mating durations (in minutes) for a single individual, with each dot corresponding to a single mating event. Statistical details are given in [Figure 1-table supplement 2](#).

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Reviewer #1 (Public review):

Summary:

This study asks how selection for male aggressiveness affects life-history and reproductive fitness traits in *Drosophila melanogaster* males.

Strengths:

Multiple comprehensive assays are used to address the question.

Weaknesses:

(1) The flies used for comparisons are inadequate. Behavioral assays compare Bully males mated to non-coevolved Cs females with Cs males mated to coevolved Cs females.

(2) Lifespan analysis is done on male progeny of Cs females mated to either genetically more distant Bully or co-evolved Cs males; the longer lifespan and performance on the former is interpreted as a trade-off with aggressiveness, rather than a simple explanation of hybrid vigor.

(3) Differences in CHCs between Bully and Cs males and Cs females mated to those males are not shown to cause differences in measured behavioral outcomes.

<https://doi.org/10.7554/eLife.107559.1.sa2>

Reviewer #2 (Public review):

Summary:

The authors compare "Bully" lines, selected for male aggression, to Canton-S controls and find that Bully males have lower mating success, shorter mating durations, and remate sooner. Chemical analyses show Bully males have distinct cuticular hydrocarbons (CHC) signatures and transfer markedly less cVA to females, offering a plausible mechanistic link to weaker mate-guarding.

Paradoxically, Bully males live longer and remain fertile at older ages when CS males no longer mate, indicating a shift in the reproduction-survival trade-off in aggression-selected populations.

Importantly, the work sheds light on proximate mechanisms, demonstrating that shifts in CHCs and pheromone transfer co-occur with changes in fitness traits, thus offering new entry points for understanding life-history evolution.

Strengths:

The manuscript's strengths lie in its comprehensive and integrative approach framed within an evolutionary context. By combining behavioral assays, chemical profiling, and lifespan measurements, the authors reveal a coherent pattern linking aggression selection to life-history trade-offs. The direct quantification of cVA in female reproductive tracts after mating provides a particularly compelling mechanistic correlate, strengthening the link between behavior and chemical signaling. Findings on altered 5-T and 5-P levels further highlight how chemical communication shapes mating and mate-guarding strategies. Analytical approaches

are largely rigorous, and the results provide valuable insights into the pleiotropic effects of selection on socially relevant traits. The study will be of interest to *Drosophila* biologists working on sexual selection, behavioral evolution, and aging.

Weaknesses:

The weaknesses are primarily conceptual rather than procedural. The generality of the findings is uncertain, as selection appears to be represented by only one (and a second closely related) Bully line, limiting conclusions about selection responses versus line-specific drift or founder effects. The causal link between aggression selection and increased longevity is not established: the data show a correlated shift but do not identify mechanisms underlying lifespan extension. In several places, the manuscript uses causal language (e.g., that selection 'influences' longevity or mating strategy) where association would be more accurate; this should be toned down to avoid overstatement. Ecological relevance is also not addressed, since laboratory conditions may bias the balance between costs and benefits of aggression compared with variable natural environments. Addressing these points would strengthen both the impact and clarity of the study.

<https://doi.org/10.7554/eLife.107559.1.sa1>

Author response:

eLife Assessment

*This valuable study addresses the effects of selection on aggression on fitness and life-history trade-offs in *Drosophila melanogaster*. However, the evidence presented is incomplete and does not support the claims proposed in the study of increased survival of highly aggressive males at the expense of reproductive success and shorter mating duration. The main limitation of the study is the choice to use males from only one aggressive *Drosophila* line in combination with CantonS females, that do not allow disambiguation between nonaggression-related factors, such as hybrid vigor and aggression-related factors influencing mating and lifespan.*

We would like to clarify the points raised in the eLife assessment.

The report states that we relied on a single line of hyper-aggressive males tested with CantonS females, and implies that Bully and Cs have not co-evolved. This is a misunderstanding: Bully flies were derived from Cs population. Thus, Bully and Cs have co-evolved. In addition to the Bully A line presented in the main figures of the manuscript, we replicated several of our findings with a second independent selected line, Bully B. Results from courtship assays involving both Bully A and Bully B couples males and females were presented in Figure Supp1. We apologize for not having made this more explicit in the original manuscript, which we will correct. These experiments should alleviate the concerns from the reviewers; they demonstrate that our conclusions are supported by two independent hyper-aggressive lines, and these include assays with selected male and female flies.

Public Reviews:

Reviewer #1 (Public review):

Summary:

*This study asks how selection for male aggressiveness affects life-history and reproductive fitness traits in *Drosophila melanogaster* males.*

Strengths:

Multiple comprehensive assays are used to address the question.

We thank the reviewer for recognizing these strengths.

Weaknesses:

(1) The flies used for comparisons are inadequate. Behavioral assays compare Bully males mated to non-coevolved Cs females with Cs males mated to coevolved Cs females.

We thank the reviewer for this comment, which made us realize that we had not sufficiently highlighted some of our experiments. The Bully lines used in our work were derived from Canton-S flies and thus did co-evolve with Cs. As originally described by Penn et al. (2010), highly aggressive “Bully” lines were generated through selective breeding from Canton-S males that consistently won aggressive encounters. After 34–37 generations, stable Bully lines were established. Thus, Bully and Cs flies have co-evolved and 2) the selection applied was male-specific. Independent selection replicates produced distinct lines, including Bully A and Bully B. Previous studies only characterized Bully A (Penn et al., 2010; Chowdhury et al., 2017), but our work includes both Bully A and Bully B (Fig. S1).

The rationale for pairing Bully or Cs males with Cs females (with which both male types co-evolved) follows the approach used by Dierick et al. (2006), who investigated how the male-specific selection for aggression affected courtship and mating behaviors by testing them with standard Canton-S females. This design allows to isolate the effects of male genotype and behavior on courtship and mating outcomes, avoiding confounding effects from female behavioral changes.

We initially compared selected Bully pairs (Bully males × Bully females) (Fig. S1) with Cs pairs and observed similarly shortened mating durations in both Bully × Bully and Bully × Cs matings (Fig. S1, Fig. 1F and G). Thus, the reduction in mating duration arises specifically from Bully males. We therefore chose to use Cs females as a standard background to assess the consequences of male-specific selection for aggression on reproductive behaviors.

(2) Lifespan analysis is done on male progeny of Cs females mated to either genetically more distant Bully or co-evolved Cs males; the longer lifespan and performance on the former is interpreted as a trade-off with aggressiveness, rather than a simple explanation of hybrid vigor.

We appreciate this comment, which again stems from a poor explanation from our part about the origin of the Bully line in the original manuscript. The Bully flies were derived from the same original population as the Cs line. Hybrid vigor typically arises when crossing individuals from distinct populations, which is not the case here as both Bully and CS come from the same population.

To further support our conclusions, we conducted additional experiments using progeny from within-line crosses (Bully males × Bully females) and results revealed the same phenotype: the progeny of these flies also exhibited significantly longer lifespans than Cs males × Cs females progeny. This finding argues against hybrid vigor as the main explanation for the observed phenotype, since both the Bully and Cs crosses result in inbreeding, yet give longer lifespan in Bully. We will include these additional longevity data (currently not included in the manuscript) to strengthen our results and reinforce our interpretation.

(3) Differences in CHCs between Bully and Cs males and Cs females mated to those males are not shown to cause differences in measured behavioral outcomes.

We thank the reviewer for raising this important point regarding causality. One way to establish a causal link between differences in CHCs observed in Bully and Cs flies and the corresponding behavioral outcomes would be to experimentally manipulate CHC profiles. For instance, one could perfume oenocyte-less males with the compounds found in higher abundance in Bully flies, then perform behavioral assays to assess causality. We agree that such experiments would be highly informative in determining the functional roles of specific CHCs elevated in Bully males. However, this approach is technically challenging, as the perfuming technique must be optimized to transfer precise amounts of each compound. For example, this method can be used to gradually perfume flies to assess dose–response behavioral effects, whereas matching exactly the natural concentrations found in individuals, especially given inter-individual variability, remains difficult.

We considered conducting such experiments during our study but did not pursue them for these technical reasons. Nevertheless, we can include a statement in the Discussion acknowledging this as an important future direction to test the causal relationship between CHC variation and behavior.

Reviewer #2 (Public review):

Summary:

The authors compare "Bully" lines, selected for male aggression, to Canton-S controls and find that Bully males have lower mating success, shorter mating durations, and remate sooner. Chemical analyses show Bully males have distinct cuticular hydrocarbons (CHC) signatures and transfer markedly less cVA to females, offering a plausible mechanistic link to weaker mate-guarding.

Paradoxically, Bully males live longer and remain fertile at older ages when CS males no longer mate, indicating a shift in the reproduction-survival trade-off in aggression-selected populations.

Importantly, the work sheds light on proximate mechanisms, demonstrating that shifts in CHCs and pheromone transfer co-occur with changes in fitness traits, thus offering new entry points for understanding life-history evolution.

We thank the reviewer for this positive summary of our work.

Strengths:

The manuscript's strengths lie in its comprehensive and integrative approach framed within an evolutionary context. By combining behavioral assays, chemical profiling, and lifespan measurements, the authors reveal a coherent pattern linking aggression selection to life-history trade-offs. The direct quantification of cVA in female reproductive tracts after mating provides a particularly compelling mechanistic correlate, strengthening the link between behavior and chemical signaling. Findings on altered 5-T and 5-P levels further highlight how chemical communication shapes mating and mate-guarding strategies. Analytical approaches are largely rigorous, and the results provide valuable insights into the pleiotropic effects of selection on socially relevant traits. The study will be of interest to Drosophila biologists working on sexual selection, behavioral evolution, and aging.

We thank the reviewer for recognizing the integrative design and mechanistic contributions of our study.

Weaknesses:

The weaknesses are primarily conceptual rather than procedural. The generality of the findings is uncertain, as selection appears to be represented by only one (and a second closely related) Bully line, limiting conclusions about selection responses versus line-specific drift or founder effects. The causal link between aggression selection and increased longevity is not established: the data show a correlated shift but do not identify mechanisms underlying lifespan extension. In several places, the manuscript uses causal language (e.g., that selection 'influences' longevity or mating strategy) where association would be more accurate; this should be toned down to avoid overstatement. Ecological relevance is also not addressed, since laboratory conditions may bias the balance between costs and benefits of aggression compared with variable natural environments. Addressing these points would strengthen both the impact and clarity of the study.

(1) Generality of findings and potential line effects

We agree that our results presented in the main figures of the manuscript relied mainly on one Bully line (Bully A). To address potential line-specific effects, we replicated key courtship experiments with another independent line, Bully B, selected in parallel from the same Canton-S stock but through distinct selection replicates. The results obtained from Bully B closely matched those from Bully A, suggesting that the observed phenotypes are consistent consequences of aggression selection rather than random drift or founder effects.

(2) Causality versus correlation

We concur that some sentences in the manuscript could overstate causal interpretations. We will revise the text to clearly distinguish correlation from causation and to avoid implying direct causal relationships where data only support association.

(3) Ecological relevance

We appreciate this point. Our experiments were performed under controlled laboratory conditions, which may not fully capture the ecological contexts shaping the costs and benefits of aggression. We will acknowledge this limitation and expand the Discussion to consider how environmental variability could modulate the fitness trade-offs associated with aggression in natural populations.

We thank both reviewers for their constructive feedback, which will help us strengthen the rigor and clarity of the manuscript. We believe that the additional results and revisions will satisfactorily address their concerns.

<https://doi.org/10.7554/eLife.107559.1.sa0>